

EXERCISE SHEET 1

This is not the official exercise sheet from the course.

More difficult exercises are denoted by ★

Exercise 1.

Let the universe be the rational numbers $\mathbb{Q}$.

We define the following predicates

$$P(x) = \begin{cases} 1, & x \in \mathbb{Z} \\ 0, & \text{Else} \end{cases} \quad Q(x) = \begin{cases} 1, & x > 2 \\ 0, & \text{Else} \end{cases} \quad R(x, y, z) = \begin{cases} 1, & x^3 + y^3 = z^3 \\ 0, & \text{Else} \end{cases}$$

In this context, describe the following statements in words. Decide for each statements if its true or false.

1. $\exists x(P(x) \wedge Q(x))$
2. $\exists x(Q(x) \wedge \neg P(x))$
3. $\forall x(Q(x) \rightarrow P(x))$
4. $\forall x \forall y \exists z(R(x, y, z))$
5. $\exists x \exists y \exists z(P(x) \wedge P(y) \wedge R(x, y, z))$
6. ★ $\neg \exists x \exists y \exists z(P(x) \wedge P(y) \wedge P(z) \wedge R(x, y, z))$
7. ★ $\forall x \forall y \forall z((P(x) \wedge P(y) \wedge P(z) \wedge R(x, y, z)) \rightarrow (x = 0 \vee y = 0 \vee z = 0))$

Hint: For 6 and 7, look up Fermat's Last Theorem.

Exercise 2.

Let the universe be the integers $\mathbb{Z}$. Describe the following mathematical statements using predicate logic. Define your own suitable predicates and decide if the statement is true.

1. For every integer x , there exists a number y such that $x + y = 0$
2. There exists no largest natural number.
3. The sum of two odd numbers is even.
4. The product of any two numbers is strictly positive.
5. Any prime bigger than 5, is of the form $6n + 1$.
6. There exists an integer x such that for every integer y , $x \leq y$.
7. Every positive integer has an integer square root.
8. Every even integer can be expressed as the difference of two odd integers.
9. Every positive integer greater than 1 can be written as the sum of two prime numbers.
10. ★ There are infinitely many perfect squares.

Exercise 3.

Decide for each of the following claims whether it is logically valid. If it is valid, briefly justify why. If it is false, provide a counterexample using the universe of integers $\mathbb{Z}$.

1. $\forall x(P(x) \vee Q(x)) \equiv (\forall xP(x) \vee \forall xQ(x))$
2. $\exists x(P(x) \wedge Q(x)) \models (\exists xP(x) \wedge \exists xQ(x))$
3. $\exists y\forall x P(x, y) \models \forall x\exists y P(x, y)$
4. $\forall y\exists x P(x, y) \models \exists x\forall y P(x, y)$
5. ★ $(\forall xP(x) \rightarrow \forall xQ(x)) \models \forall x(P(x) \rightarrow Q(x))$

★ Exercise 4.

Let the universe U be the set of all students taking an exam.

Let the predicate Q be defined as

$$Q(x) = \begin{cases} 1, & \text{If student } x \text{ passes the exam} \\ 0, & \text{Else} \end{cases}$$

Describe the following formula in natural language and determine if it is a tautology.

$$\exists x(Q(x) \rightarrow \forall yQ(y))$$